CSE 150: Introduction to Computer Networks

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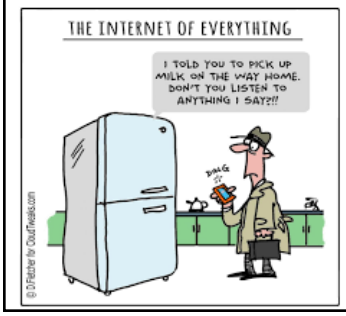
Chapter 1 slides Part 2



1

Current Assignments

- Reading:
  - Chapter 1 complete
  - By next Tuesday:
    - Read Chapter 2 (2.1, 2.2, 2.4)
- Lab1 due Saturday
- Group Assignment 1 due Sunday
- Homework:
  - R problems in Canvas module
  - 2 pages in this slide deck "Homework" as we reach them

12

12

Upcoming Events

- Tutorials with student tutors
  - Schedule on Canvas
  - All week
  - Get your Python help!

13

13

Chapter 1: Roadmap – sections of text

1.1 ✔ What *is* the Internet?

1.2 ✔ network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, Internet structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

Note: Section #s according to 7th Edition

15

15

Fundamental Assumption

- Networks are shared
  - Routers and links

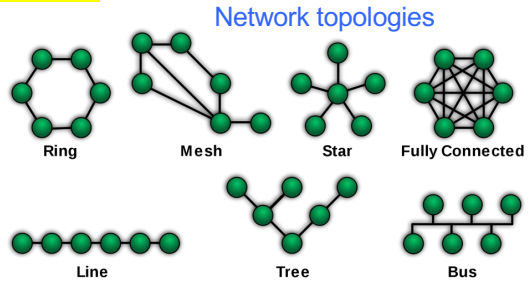
16

16

Organizing the nodes: Dedicated- vs shared

Cost-benefit trade-off arrangement of links: delay, cost, scalability and robustness

Network topologies



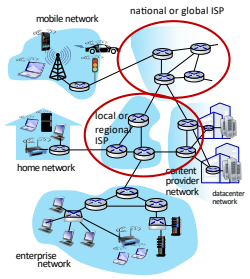
From Wikipedia

Multiple users share the links (except … ?

17

2 Approaches to Sharing a Communication Path

How to move data packets through the network?



1. Reserved – Circuit Switching
   - Make arrangements ahead of time
   - Roots in the telephone network
2. On-demand - Packet Switching
   - Use whenever you want

- Think about a restaurant and # customers served

18

Chapter 1: roadmap and readings

1.1 ✓ what *is* the Internet?
1.2 ✓ network edge
- ✔ end systems, access networks, physical media

1.3 network core
- packet switching,
- circuit switching
- Internet structure – network of networks

1.4 delay, loss, throughput in networks
1.5 protocol layers, service models
1.6 networks under attack: security
1.7 history

21

Circuit Switching
(reserved use of network)

- Roots in the telephone network
- Resources (buffers, link transmission rate) along path are "**reserved**" ahead of time
- All data in the connection flows over this "circuit"
- Service guarantee: user receives the service that was guaranteed
  - e.g., guaranteed transmission rate
- Not the method in use in the Internet

22

Early Telephone Network (POTS)

Physical circuit set up between caller and "callee"
- Pt to pt links→not scalable
- Giant wooden switchboard " patch panel"

~ 1903

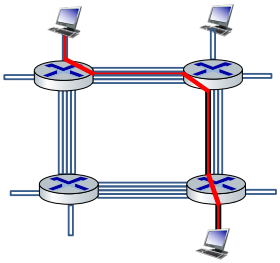
PSTN (Public Switched Telephone Network)
- Circuit Switching technique
  - Customer buys fixed capacity for the duration of calls
    - Call is established before talking
    - data rate and price e.g. 64 kbps, 5-minute call
    - Line quality "guaranteed"
- Is there sharing?
  - Yes, but by reservation, one at a time

23

Circuit switching in a network

Homework: Numerical example in Section 1.3.2 in book: (640,000 bits from A→B)

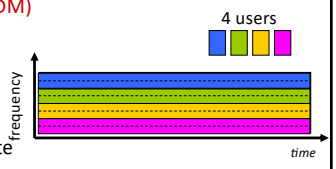
End-end resources allocated and reserved for "call" between source and destination
- Dedicated resources:
  - circuit-like (guaranteed) performance
- Each link has four circuits:
  - A "call" (red) gets 2nd circuit in top link and 1st circuit in right link.
  - All data takes established path
  - Service guarantee: some fraction of the overall bandwidth
- Takes time to set up the circuit
- Idle: when circuit is not used (no data flow), the circuit is "idle"
  - Re-claiming unused (idle) resources NOT possible until circuit is torn down



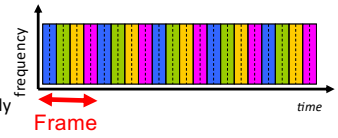
24

Channel Partitioning: FDM and TDM

Frequency Division Multiplexing (FDM)
- Optical & electromagnetic frequencies divided into narrow frequency bands
- Each user allocated own band
- Within their narrow band, users transmit continuously at max rate possible
  - User rate < total link capacity



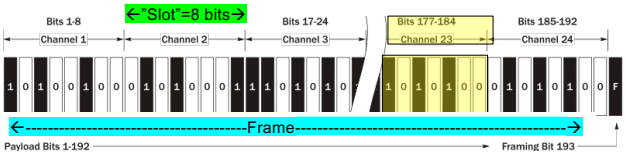
Time Division Multiplexing (TDM)
- A frame of data is divided into Time Slots (AKA channels)
- Users allocated Time Slot
- Users transmit at full link rate, but only during Time Slot
  - i.e. not continuous transmission



26

Using TDM to Guarantee Bandwidth

The setup: 1.536Mbps link
24 users want service



- Bandwidth guarantee: 1.536Mbps/24 = 64kbps per user

- Frame divided into 24 channels
- Each user sends 8 bits per channel (= ? bits/frame)
  - 8000 frames/sec → System accepts 8 bits from each user every 125 microseconds
  - bandwidth guarantee
- Play around with the numbers and verify
  - Convince yourself that 8 bits per user per frame achieves bandwidth guarantee
- What happens during **silent periods**? Think about link utilization.

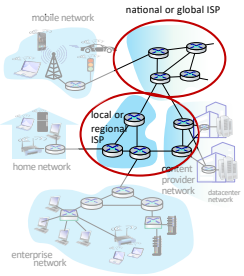
27

Packet Switching

29

Internet Core – Packet Switching

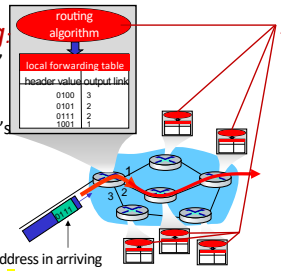
- Packet-switching: Application-layer messages broken into smaller *packets*
  - Independent unit of transmission
  - Transmitted across links on path from source to destination
  - Hop by hop
- Key Features:
  - Packet can be transmitted whenever it becomes available
    - No "setup" required
  - Each packet transmitted independently over the links
    - When transmitting, full link capacity
  - Store and Forward



30

Two key network-core functions

1.Forwarding:
AKA "switching"
- *local action*– transfer arriving packets from router's input link to appropriate output link



destination IP Address in arriving packet's header.

What is a header?

2. Routing:
- *global action*-- determine source-destination paths taken by packets
- Dynamic routing algorithms

31

IP Header (network layer)

- Header is not "the data"
- envelope

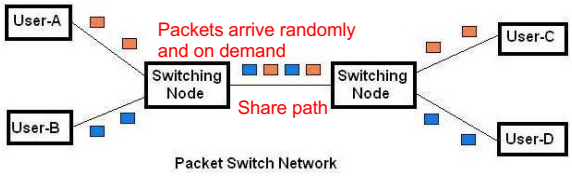
32

You see this with Traceroute

34

35

Packet Switching – Internet model



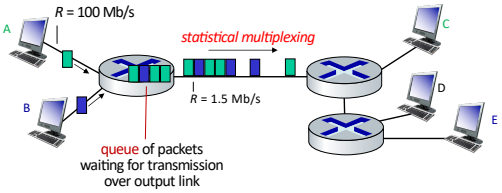
From: rfwireless-world.com

- On demand use of network, no reservations
- No service guarantees (throughput, delay, path)
- Network does its best to absorb and deliver packets to Destination
  - Packets co-exist in the network as best as possible
    - make their way from SRC to DST
  - Full bandwidth available to packets as they are transmitted over a link

Key Feature: If B has nothing to send, A can use all the bandwidth!

38

Packet Switching: Statistical Multiplexing

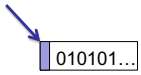


- On demand network use
- Packets arrive into network probabilistically (some distribution), no fixed pattern, as data arrives into the network
- Router and outgoing link shared by all packets
- Routers forward packets, one by one, over their outgoing links
  - "**on demand**" or "upon arrival"
  - Might have to wait!

39

Packet Switching

- How is each individual packet "standalone" ?
- Packet: header + payload



- What goes in header?
  - Each protocol has its own header →check out TCP/UDP/IP
    - Packet header may identify sender and receiver, sequence number to determine order, checksum etc.
  - IP Forwarding: destination IP Address in IP header
- Overhead cost associated with each header
  - Why is it overhead?

40

Packet Switching: Funny stuff can happen!

Tradeoff: keeping the Queue full?
Cost/benefit?

41

Are Users Happy?

43

Network Performance Metrics

- Delay
- Throughput
- Reliability

- Bit: smallest unit of data that exchanged between transmitter/receiver pairs
  - Modem gets it ready
- Byte: collection of 8 bits

44

44

Delay: Time for packets to travel to destination

- Measurement
  - Unit of time
  - Measured in seconds, *msec* or *µsec*
- Define endpoints of delay measurement
  - End-to-end
  - Link
  - Node
- Different types of Delay
  - Transmission
  - Queueing
  - Processing
  - Propagation

45

Transmission Delay: "escape" from the NIC

Host Application program produces a message with data

- Message broken into smaller chunks → *packets*
- Packet length *L* bits (Kbits, Mbits)
- Packets transmitted one at a time

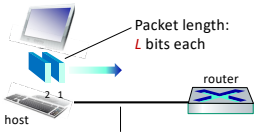
Host transmits data from interface (NIC) onto physical link – takes time!

How long does it take to push out one entire packet?

- Host transmits packet into link at *transmission rate R bits/sec*

  --*AKA link capacity, link bandwidth (Kbps, Mbps)*

Transmission delay = time needed to transmit ONE *L*-bit packet into link = $\frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$



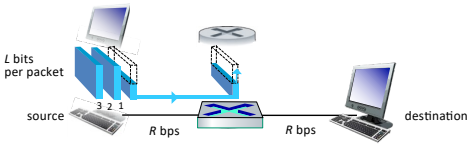
R: bits/sec transmission rate

Note: assume 0 propagation delay and 0 processing time

(think about driving somewhere: t = D / Speed)

46

Packet-switching: Store-and-Forward

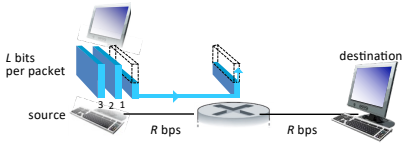


- Entire packet must be received before Router forwards over next output link
  - Router processes packets one at a time
- End-end delay for 1 packet = $2L/R$
  - 2 hops
  - (Assumption: 0 propagation delay and no processing delay in router – coming next!)
- When can the 2nd packet be transmitted?

(Note: Store-and-Forward not needed in Circuit Switching)

47

Homework –Calculator Check!



$K = 10^3$ $M = 10^6$

- What is the delay to send one packet from the Source to the destination?
- What is the total delay for 2 packets to be transmitted and received?

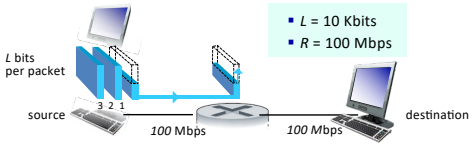
Two-hop numerical example:

- *L* = 10 Kbits
- *R* = 100 Mbps
- two-hop transmission delay = 0.2 msec
- Verify!!

Note: assume 0 propagation delay and 0 processing time

$K = 10^3$ $M = 10^6$

48



- *L* = 10 Kbits
- *R* = 100 Mbps

a) Delay for one packet transmitted from the Source to the destination?

b) What is the total delay for 2 packets to be transmitted and received?

49

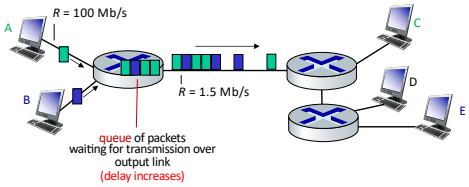
Packet-switching: Queueing

Queueing occurs when work arrives faster than it can be serviced

Remember: Store and Forward in the routers

51

How does Queue build?



Packet queuing:

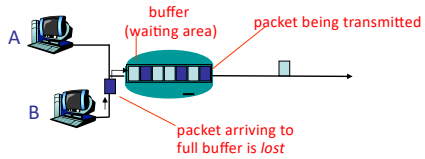
- Function of network conditions
- Arrival rate to link exceeds transmission rate for some period of time: *Mismatch:* $\lambda_{in} > \lambda_{out}$
  - → D_Q: Queueing delay! (more soon)
  - → Wait for turn!
- Queueing is expected in a packet-switched network!

52

And then… Packet Loss

- Router: Outgoing links have a Queue with finite capacity
  - Cannot increase on the fly
- Queue fills → delay increases →no space for arriving packets
- One Choice: drop a packet
- Dropped ("lost") packets may be:
  - Retransmitted previous node on path (link)
  - Retransmitted by Sender (source)
  - Ignored (nothing done)

Congestion



54

Packet Loss/ Queue Management

RED Gateways -- https://www.nytimes.com/2019/09/04/science/sally-floyd-dead.html

55

Let's Compare

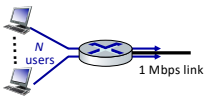
62

Circuit-switching vs. Packet switching: # users supported

This is a HW problem
(for those with course in probability only)

Example:

- 1 Mbps link
- Each user:
  - 100 Kbps when "active"
  - active 10% of time

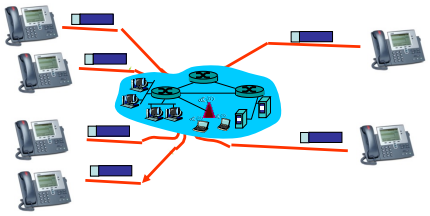


- Circuit-switching: How many users supported @100kbps?
- Packet switching:
  - With << 10 users, packets flow through system with essentially no queueing
  - With 35 users, P(# active users >10) < .004 (see HW problem in book)
  - P(<10)= 0.996 Miniscule!
  - So, when #users > 10:

Packet switching supports more users!

63

Voice over the Internet (VoIP) -not circuit switched



Voice is digitized and sent out in packets → packetized voice stream

- Packets transmitted at available bandwidth
- Delay and jitter can be a problem
- VoIP Apps – your favorite?

64

Packet Switching vs. Circuit Switching

- Make your own list of advantages and disadvantages!

Is packet switching a "slam dunk winner"?

- Does Packet Switching always win?
  - resource sharing
  - simpler, no call setup
  - Supports more users
  - No guarantees
  - What is the cost? i.e. what can happen?
- Does Circuit Switching win?
  - Advantages?
  - Disadvantages?

65

Homework

See section 1.3.2 for 3 excellent numerical examples!

Work through them in tutoring group session or with a classmate!

66

Back to Performance

67

Chapter 1: Roadmap – sections of text

1.1 ✔What *is* the Internet?

1.2 ✔ network edge

- end systems, access networks, links

1.3 ✔ network core

- ✔ packet switching, ✔ circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

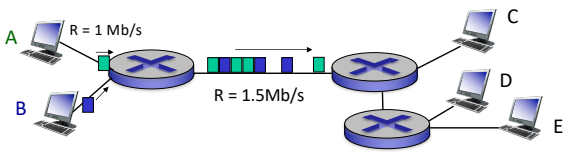
Note: Section #s according to 7th Edition

68

68

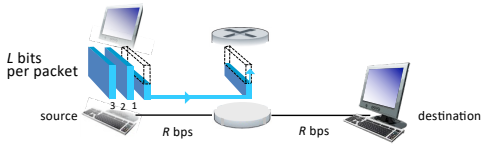
End-to-End Delay

Time it takes bits to travel from a Source to a Destination through the network.



69

Packet-switching: Store-and-Forward



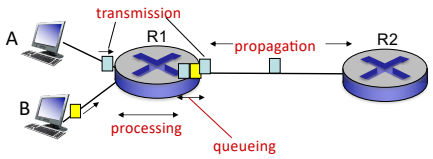
If we only consider transmission delay:

$D_{end\text{-}to\text{-}end}$ = N (L/R) for <u>one packet</u> where N = #hops (links)

But we are ignoring other delay components!
→Let's look at them!

70

Nodal Delay: 4 Sources

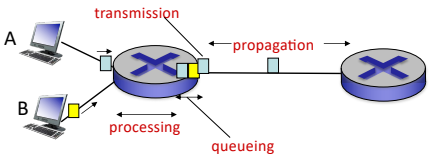


Nodal Delay: $d_{nodal} = d_{proc} + d_{queue} + d_{transm} + d_{prop}$

Processing
Queueing
Transmission
Propagation

71

Inside or Outside?



Inside Router: "<u>Service Time</u>"

D_{proc} + D_{queue}

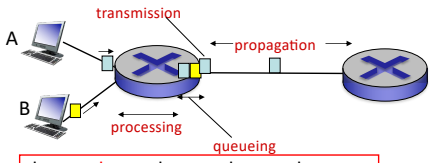
Outside Router:

D_{prop}: Propagation along link

D_{trans}: Transmission onto link

72

Nodal Delay: Processing



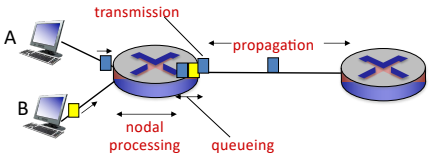
$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$

d_{proc}: <u>processing delay</u>

- check bit errors
- IP Forwarding
  - Send packet from input port, through router's switch fabric, to output port
- Policies, e.g. firewall?
- *µsec* (hardware)

73

Nodal Delay: Queueing



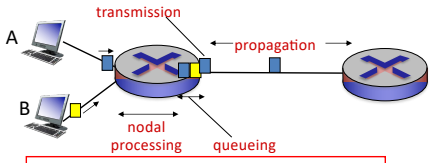
$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$

d_{queue}: <u>queueing delay</u>

- time packet waits in output queue before transmission out of node (over output link)
- Varies depending on network traffic and possible congestion in router
- *µsec* → *msec*

74

Nodal Delay: Transmission and Propagation



$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$

d_{trans}: <u>Transmission delay</u>

- L: packet length (bits)
- R: link bandwidth (bps)
- Pump bits out

d_{trans} = L/R

d_{prop}: <u>Propagation delay</u>

- d: length of physical link. (m)
- s: propagation thru physical medium (~$2x10^8$ m/sec for fiber)
- Bits travel across link

d_{prop} = d/s

Different!!

75

Journey of a Packet (length L)

76

Homework: Explore transmission delay vs. propagation delay

Applet:
https://media.pearsoncmg.com/aw/ecs_kurose_compnetwork_7/cw/content/interactiveanimations/transmission-vs-propogation-delay/transmission-propagation-delay-ch1/index.html

Goal: Examine total delay to send a packet from Source to Destination.

Keep Packet Length L=500 bytes fixed

1. Explore d_{TRANS} by keeping d=100km fixed
   a. R=10Mbps b. R=100Mbps
2. Explore d_{PROP} by keeping R=10Mbps fixed
   a. d=10km b. d=1000km

77

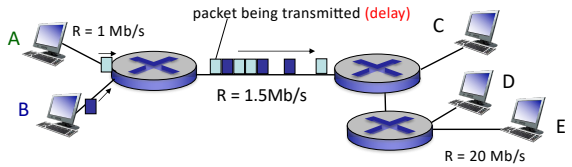
Nodal Delay: 4 Sources

- Understand the 4 sources of nodal delay
- Where is the delay located
- Practice computations
- <u>Yes, there will be computations on test</u>

79

79

End to End Delay



End-to-End Delay: Sum of all nodal delays on path. *

Variation in End-to-End Delay: What causes it?

80

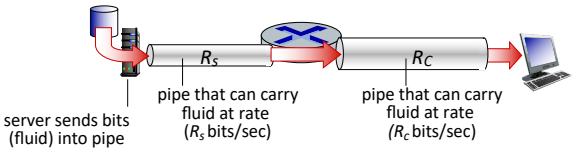
Throughput

82

82

Throughput

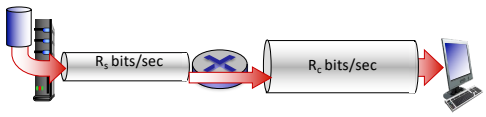
- *Throughput:* rate (bits/time unit) at which bits (transmitted by the Sender) <u>arrive at the Receiver</u>
  - *instantaneous:* rate at given point in time as seen by the Receiver
  - *average:* rate over longer period of time as seen by the Receiver → F/T where F=bits delivered and T=time interval



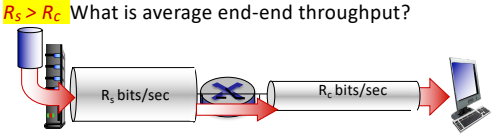
83

Throughput

$R_s < R_c$ What is average end-end throughput?



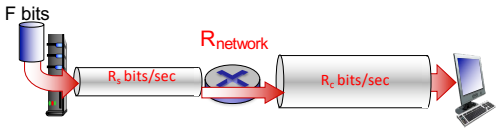
$R_s > R_c$ What is average end-end throughput?



Bottleneck link
Link on end-end path that constrains end-end throughput

84

Network usually not the Bottleneck



Upper bound on throughput:
- Maximum throughput = $\min(R_s, R_c, R_{network})$

What about delay?
- Bottleneck sets the lower bound on "distribution" time
  - Time to distribute entire file of F bits
- Distribution time = amount of data / transfer rate
  = F bits / $\min(R_s, R_c, R_{network})$

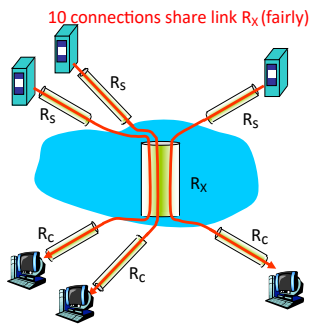
85

Bottleneck: Can be a Shared Link!

(not always just the slowest link)

- In practice: R_c or R_s is often bottleneck, but not always!
- high capacity link can become the bottleneck thru "sharing"
- Per-connection end-end throughput = $\min\{R_c, R_s, R_X/10\}$

Distribution Time:
>= $F/\min(Rs, Rc, R_X/10)$



What if one server/client pair stops transmitting?

86

Distribution Time

87

Chapter 1: Roadmap – sections of text

1.1 ✔What *is* the Internet?
1.2 ✔ network edge
- end systems, access networks, links

1.3 ✔ network core
- ✔ packet switching, ✔ circuit switching
- network structure

1.4 ✔ delay, ✔ loss, ✔ throughput in networks
1.5 protocol layers, service models

Note: Section #s according to 7th Edition

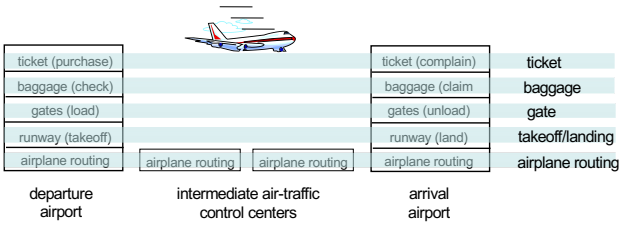
90

Layering

- The protocols in use in the Internet are layered

91

Layering of airline system



Layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

Easier to drive? ☺

92

Why layering?

- Describe a complex system
  - Break down the functional components
- Modularization eases maintenance, updating of system
  - Interface stays the same
  - Implementation can be changed, layer's service preserved → transparent to rest of system
    - e.g., suitcase check
- Cost?
  - Duplication of functionality
  - Overhead

93

Complex Internet

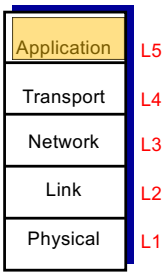
- Apps, Protocols, Switches, heterogeneous networks, End Systems and Links
- Organize network
- Put protocols in Layers
  - Provide service to layer above
  - Rely on service from layer below
- Build system by building the layers

94

TCP/IP Protocol Stack

Familiarize yourself with the layer names and function

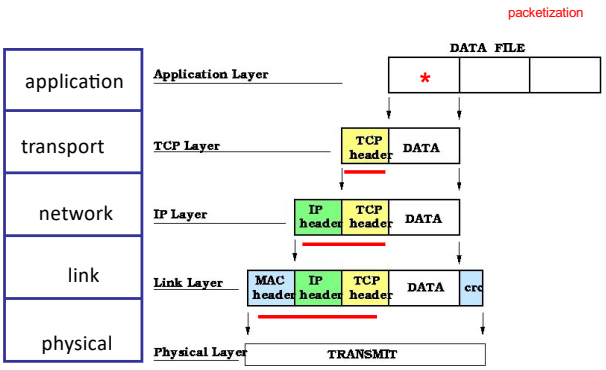
- *Application:* direct support for network applications -- HTTP, FTP, SMTP, DNS
- *Transport:* Process-to-process communication, reliable/unreliable data transfer -- TCP, UDP
- *Network:* Host-to-Host delivery of IP datagrams (packets) from source host to destination host (IP, routing protocols)
- *Link:* Data transfer between End Systems connected by a single physical link
  – Ethernet, 802.11 (WiFi)
- *Physical:* individual bits "on the wire"



L4, L3 and part of L2 in Operating System

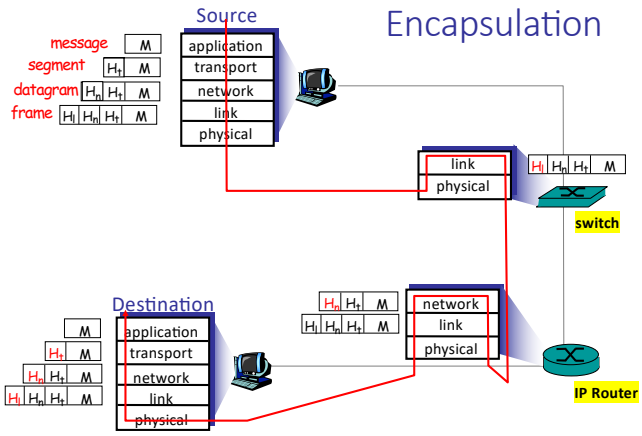
95

Encapsulation: payload and the cost



96

Encapsulation

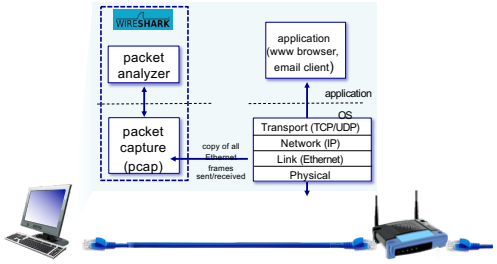


97

Decapsulation

- Frame arrives and accepted by NIC
- Up the protocol stack ….. Message delivered to Application Layer

Wireshark



App Messages, IP and Transport layer headers